

EXECUTIVE SUMMARY

# What The Backtest Is Actually Saying

This is a short decision memo, not the full report. It separates the investable core from the eye-catching but hard-to-survive ideas.

Generated 2026-05-10 15:27 UTC | Current regime: Fragile Late-Cycle Watch  
| Quadrant: Growth Down / Inflation Down | History tested: 2006-2026.

PRINCIPLE 1

## Protect The Downside First

The reserve rule earns less than the fastest strategy, but it keeps the path smoother. That matters because strategies only work if you can hold them.

MEDIUM RISK / LOWER EFFICIENCY

## Reserve Rule

This earns a return, but the drawdown path is harsher than the reward profile deserves.  
CAGR +6.3% | Max DD -46.7%

PRINCIPLE 2

## Borrow Only After You Earn The Right

Leverage is not alpha. It amplifies whatever quality already exists. Here it raises reward and cuts survivability.

MEDIUM RISK / LOWER EFFICIENCY

## Quality Rotation

This earns a return, but the drawdown path is harsher than the reward profile deserves.  
CAGR +6.1% | Max DD -51.4%

BOTTOM LINE

# Keep The Unlevered Reserve Rule

- Best unlevered growth engine: Quality Rotation at about +6.1% CAGR with -51.4% max drawdown.
- Best steadier live profile: Reserve Rule at about +6.3% CAGR with -46.7% max drawdown.
- The 2000% number is total return in the leveraged ML path, not an annual rate.
- SPY x3 shows why leverage alone is dangerous: +8.7% CAGR with -92.4% max drawdown.

PRINCIPLE 3

## Separate Prediction From Proof

The model signal is one thing. The proof is the untouched holdout and the walk-forward history. They are separate on purpose.

EXTREME RISK / WEAK REWARD QUALITY

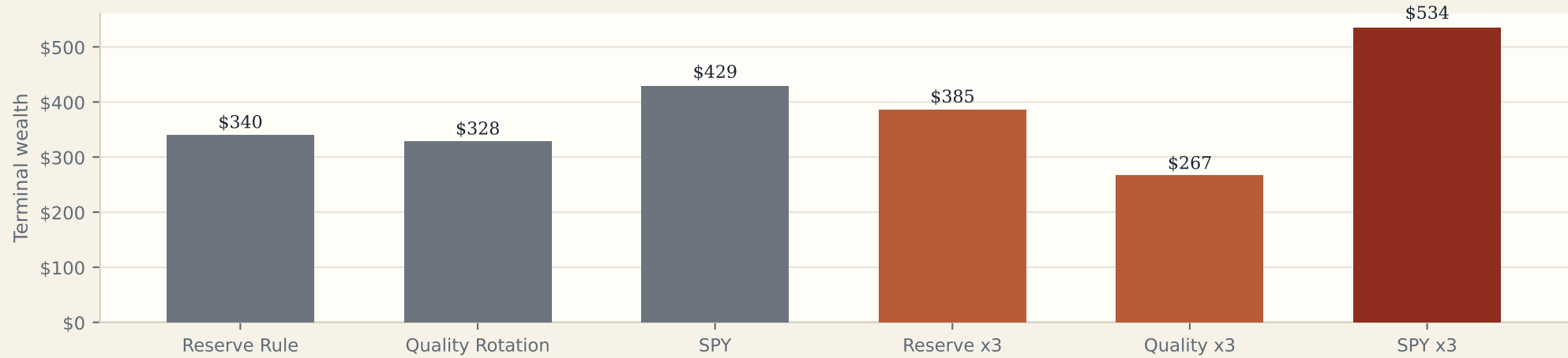
## SPY x3

Borrowing magnifies the pain faster than it improves the compounding path.  
CAGR +8.7% | Max DD -92.4%

## Risk / Reward Map



Terminal Wealth From \$100



This is why the leveraged rows show 2000%+ total return while still carrying severe path risk.

RISK LADDER

High Reward Is Not The Same As High Quality

| Strategy         | Bucket                         | Hist CAGR | Hist DD | Holdout CAGR | Holdout DD |
|------------------|--------------------------------|-----------|---------|--------------|------------|
| Reserve Rule     | Medium risk / lower efficiency | +6.3%     | -46.7%  | +9.1%        | -0.9%      |
| Quality Rotation | Medium risk / lower efficiency | +6.1%     | -51.4%  | +13.6%       | -1.4%      |
| SPY              | Medium risk / lower efficiency | +7.5%     | -49.3%  | +11.0%       | -2.3%      |
| Reserve x3       | High risk / high reward        | +6.9%     | -68.4%  | +9.7%        | -1.0%      |
| Quality x3       | High risk / high reward        | +5.0%     | -93.2%  | +38.7%       | -4.4%      |
| SPY x3           | Extreme risk / weak reward     | +8.7%     | -92.4%  | +29.9%       | -7.2%      |

WHY THIS IS NOT FORWARD-LOOKING

Controls Against Bias

- Features are lagged 1 bar before training and prediction.
- Each fold uses a 5-bar purge and 5-bar embargo.
- The final holdout starts on 2025-01-01 and is never used for model selection.
- Signals execute one bar later and hold 1.5 bars.

Decision

What To Stick To

- Core portfolio: unlevered reserve cash rule.
- Growth sleeve: unlevered quality rotation if you can tolerate the deeper drawdown path.
- Tactical only: reserve x3